Abhishek Agrawal

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SUMMARY

Data Engineer with 5+ years of expertise in building scalable, resilient cloud solutions on AWS & GCP, leading migrations, and automating operations with Python, SQL, Spark, Airflow, Snowflake, & Databricks. At NBCUniversal, partnered with business owners and led teams to deliver C-suite Olympics reporting 60 mins ahead of SLA with zero breaches, cut monthly cost 30%, and turn manual workflows into one-click utilities that sped up delivery ~75%. Seeking to turn your toughest data problems into faster, cheaper, automated solutions.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Scala, PySpark, Java, Go, JavaScript, TypeScript, Shell/Bash
- **Big Data:** Apache Spark, Kafka, Flink, Airflow, Hadoop HDFS, Hive, HBase, MapReduce, Dagster, dbt, Great Expectations; ETL/ELT
- **Data Warehousing | Lakehouse:** Snowflake (Snowpipe, Secure Data Sharing), Databricks (Unity Catalog); Iceberg, Delta Lake
- **Databases:** PostgreSQL, MySQL, Microsoft SQL Server, SingleStore, Cassandra, MongoDB; Schema Design, Query Optimization, Indexing
- **Cloud:** AWS (EC2, EMR, Lambda, IAM, S3, RDS, CloudWatch, Redshift, Athena, Glue), GCP (GKE, BigQuery), Azure
- **DevOps & CI/CD:** Docker, Kubernetes, Terraform, GitHub Actions, GitLab CI, Jenkins, Datadog; Build & Deployment Automation
- **Backend & APIs:** REST APIs (Flask, FastAPI, Django), GraphQL, Microservices, API Gateway, Server-Side Rendering
- **Frontend Technologies:** React.js, Angular, HTML5, CSS3, Tailwind CSS, Bootstrap, MVC
- **BI & Visualization:** Power BI, Tableau, QuickSight, MicroStrategy
- **AI & LLM:** OpenAI, Claude (Skills, MCP Servers), Gemini, Hugging Face, Ollama, Hermes, Cursor, PyTorch, MLOps
- **Collaboration:** Jira, Confluence, Agile/Scrum, Slack, Code Reviews, SDLC, Unit Testing, Debugging & Troubleshooting

EXPERIENCE

NBCUniversal Media (Comcast Corp) — Pune, India

Data Engineer II — **Sept 2022 – Aug 2024**

Rotation 3 | C-Suite Olympics'24 Migration

- Delivered executive ad-performance reports 60 mins ahead of SLA with zero breaches or escalations through the Paris Games, on auto-scaling ETL with built-in quality checks and alerts.
- Cut monthly cost 30% and ended daily failures by leading a team of 5 to migrate 40+ on-prem jobs to Spark (Scala) on EMR Serverless with event-driven triggers.
- Absorbed a 60% scope expansion while preserving original deadlines by engineering an operationally scalable Airflow orchestration framework that now powers 30+ pipelines across linear, digital, and CTV.
- Partnered directly with BI teams and ad-sales owners to redesign the data model on an Iceberg lakehouse with Redshift, tested and deployed via CI/CD.

Rotation 2 | Alt-Currency Automation

- Automated currency-partner ingestion with zero failed runs and lower compute cost, buffering late, out-of-order inbound batches into one parallelized orchestration that stages each batch concurrently and finishes with a shared metric refresh.
- Standardized 15–20 partner pipelines (VideoAmp, EDO) exchanging millions of daily impression records, building shared schema and metric layers with a team of 7 for a single source of truth that eliminated duplication errors and sped up onboarding.
- Retired prod-parallel resources and temporary production patches by serving on-demand data requests as config-driven jobs through the same workflow.
- Shipped with zero manual deployments through event-driven ingestion (S3 → SQS → Lambda → Airflow → EMR → Redshift), dynamically generated staging tasks, shared Scala/Python packages, and CI/CD.

Data Engineer I — **Aug 2021 – Aug 2022**

Rotation 1 | Audience Insights Hub (NBCUnified · Performance Insights Hub)

- Cut pipeline delivery time ~75% with one-click, SQL-driven features for audience overlap, segment activation, and ETL, letting business teams match 150M+ household records against partner data without a data engineer.
- Scaled the "SQL in, ETL out" engine to onboard new data clean rooms after rebuilding it from scratch in a third of the original code.
- Connected siloed departments & vendors without changing infrastructure through a shared business-name catalog and zero-copy data sharing.
- Reduced compute cost up to ~20% on migrated pipelines with SLA-based cluster presets users select in the app.
- Automated CCPA/PII compliance across data clean rooms, running purge requests as parallel background jobs, while governing user access with role-based access controls (RBAC).
- Engineered full, incremental, and micro-batched loads in & out of S3, Snowflake, and Redshift, with Spark (Scala) ETL on EMR/Databricks, Snowflake-native ELT, Airflow/Step Functions orchestration, Jinja2-generated jobs and Terraform, and Flask APIs over a PostgreSQL catalog, all managed using CI/CD (GitLab).

PROJECTS

***JobSimp** · Chrome Extension (Manifest V3), JavaScript, IndexedDB, RAG, Gemini/Claude/OpenAI*

- Reduced my per-job application time from ~20 mins to a few minutes, adding 30+ applications a day with one-click autofill, automated tracking, and job filtering.
- Scored any job posting with a fit verdict, match/miss breakdown, & resume changes needed, benchmarked with LinkedIn scores.
- Tailored my resume, cover letter, and essay answers to each job description, with an interactive apply flow that handles complex form widgets across multi-page applications.
- Engineered a Manifest V3 extension with a service-worker controller over an IndexedDB property graph, using RAG retrieval and targeted page scraping to keep Gemini/Claude/OpenAI token usage low per application.

***NYC Housing Sales (2015–2025)** · NYU Coursework · Python, AWS S3, Tableau*

- Turned ~900K messy NYC property-sale records into a clean, analysis-ready lakehouse, resolving inconsistent schemas, malformed addresses, and duplicate parcels.
- Traced a decade of price-per-sqft, sales-volume, and affordability shifts in an interactive Tableau dashboard by borough, neighborhood, and price tier.

ADDITIONAL EXPERIENCE

New York University **New York, US**

Research Assistant **Dec 2025 – May 2026**

***HPC Telemetry Lakehouse & Power-Forecasting Pipeline** · Diffusion Model, NYU HPC, GCP, Python, PyTorch, PySpark, Parquet*

- Sped up processing ~5x on ~2 TB of MIT SuperCloud telemetry (460K+ jobs) by architecting a medallion lakehouse on a 16-node Spark cluster, tuned with AQE, partitioning, and skew handling.
- Enabled datacenter power forecasting before jobs run, generating GPU power traces for 4,400+ unseen jobs (peak correlation 0.93) with a sliding-window autoregressive diffusion transformer conditioned on SLURM job metadata and the two preceding windows.

Teaching Assistant **Aug 2025 – Dec 2025**

Data Engineering Course

- Designed and graded the full assessment set for 45 students: assignments, midterm, final exam, and projects.
- Guided student projects from proposal to final demo through proposal reviews, progress checks, and technical support, while managing lecture materials.

EDUCATION

New York University, US — MS, Computer Engineering; GPA: 3.5/4 **Aug 2024 – May 2026**

Nagpur University (RCOEM), IN — BS, Computer Engineering; GPA: 3.5/4 **Aug 2017 – May 2021**